

AI PROJECT

Fake News Detection



**FILTERING
FACTS
FROM
FICTION**



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project



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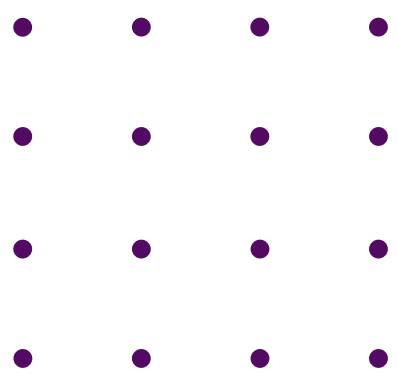
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Testing
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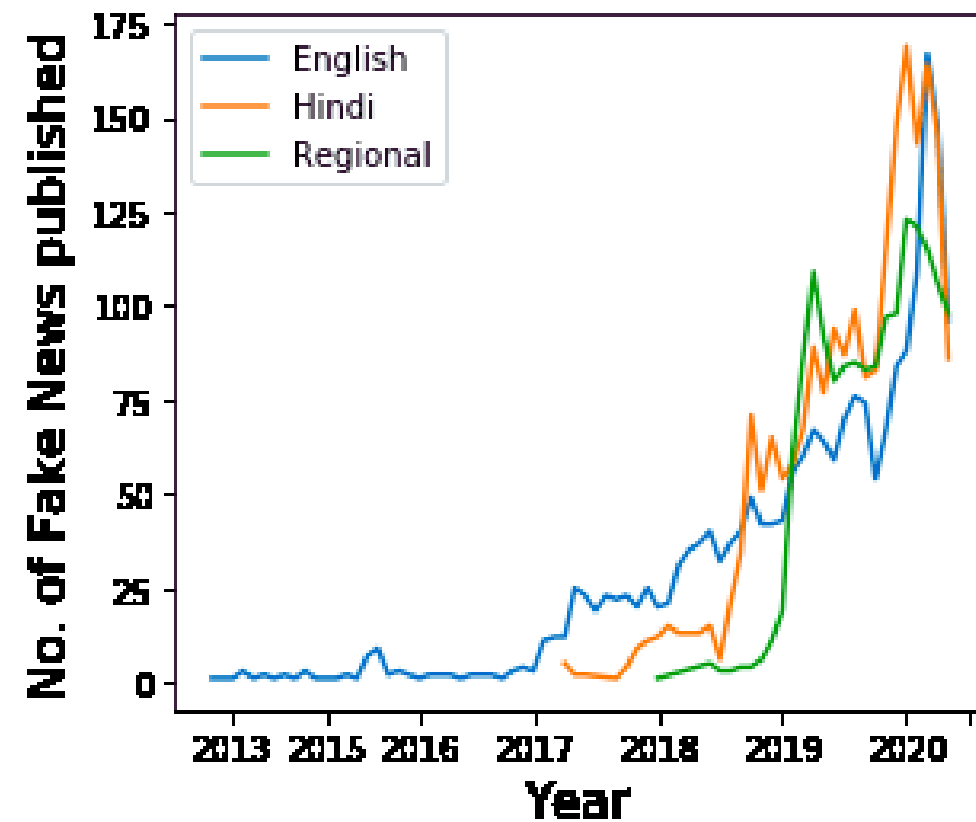
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Introduction

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“When facts speak, even the loudest lies lose their voice”



The damage caused due to fake news on social media has increased due to the growth of the internet penetration in India, which has risen from 137 million internet users in 2012 to over 600 million in 2019.

The graph illustrates the rapid year-over-year increase in the number of fake news incidents, driven by the growing rise in digital usage.

In today's digital age, the spread of fake news has become a significant threat to society, influencing public opinion, political decisions, and social harmony. The rise of social media platforms and instant information sharing has made it easier for misinformation to spread rapidly, often outpacing the truth. This has created a critical need for effective fake news detection systems to maintain the integrity of information and safeguard democratic processes.

WHY IS IT NEEDED AND How CAN AI HELP

The need for an efficient fake news detection system has never been greater. Traditional methods of fact-checking are often slow and labor-intensive, making them inadequate to combat the sheer volume and speed at which misinformation spreads. This is where Artificial Intelligence (AI) plays a transformative role. AI-powered fake news detection systems can analyze vast amounts of data in real-time, identify patterns of deception, and distinguish between reliable content and false narratives using advanced machine learning algorithms and natural language processing (NLP).



Problem Scoping

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How can we help students and community members find a way to identify and verify the accuracy of information so that they can reduce their exposure to misinformation and make better-informed decisions?

Main Issue's:

1. Misleading content
2. Data fabrication for personal benefits
3. Distrust in online sources
4. Time and effort wasted to debunk the false claims
5. Doubt in the credibility of data



Users And Solution

Students

If so many people are talking about it online, it has to be true.

Teachers

I want students to develop critical thinking skills to spot fake news.



Parents

Is my child getting a balanced perspective, or are they being misled?

Administrators

How do we maintain trust in our communications when misinformation spreads so quickly?

The best AI solution for our users and team is the "Fake News Detector." This tool will use advanced natural language processing (NLP) and machine learning algorithms to analyze news articles and identify potential misinformation. By cross-referencing content with trusted sources and detecting patterns often associated with fake news, this tool will provide users with a reliable method to verify the authenticity of information. We will implement this solution using existing tools like python for content analysis and custom-built machine learning models for fact-checking.

How fake news detection works ?

1. Users will visit the Fake News Detector
2. Users will paste the article text they want to check.
3. Users will see the option to use the submit button.
4. Users will click a "Submit" button to start the AI-powered analysis process.
5. The tool will present real for real news and fake for fake news.

Step 1:  **Paste the text** 

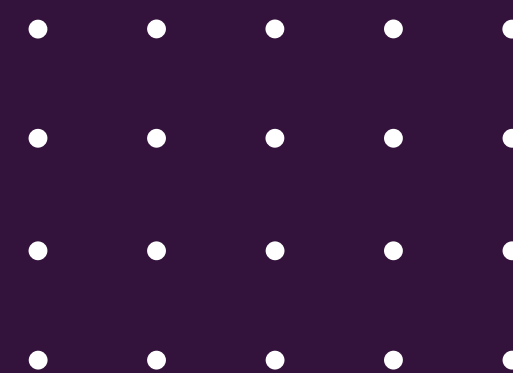
Step 2:

submit 

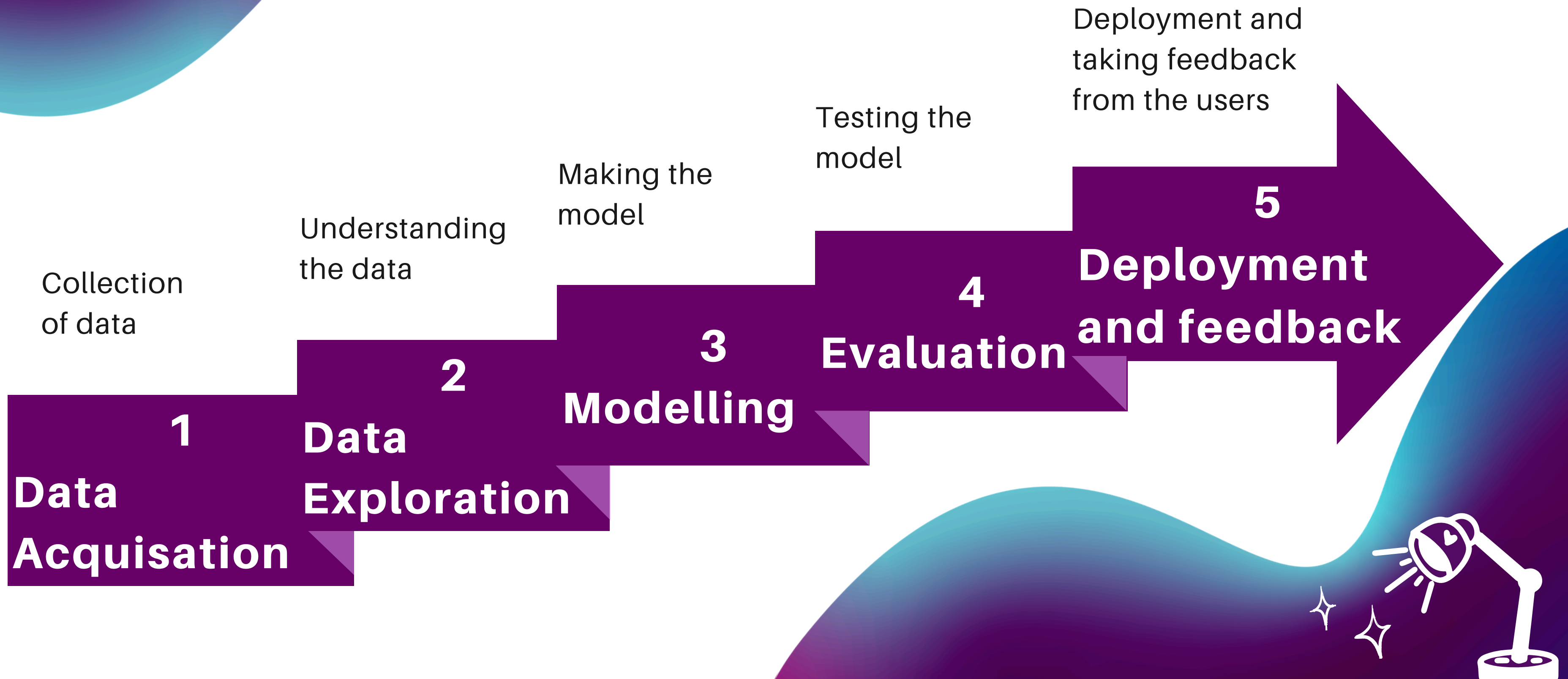
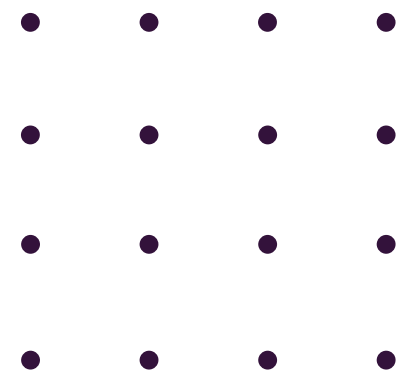
Step 3:

Real

Fake



Project flow



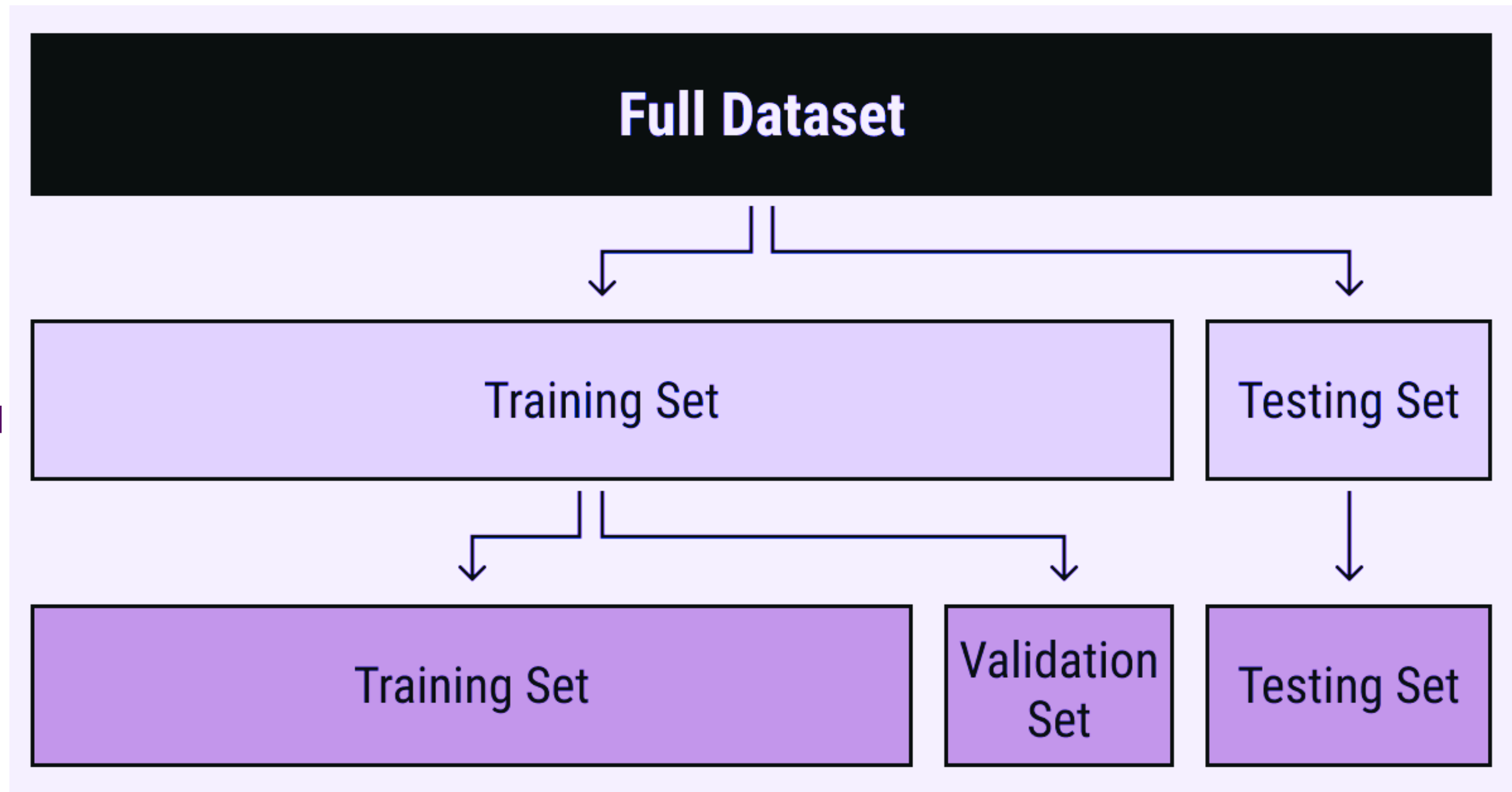
Data Acquisition

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How to use the Dataset in AI ?

Collecting data allows you to capture a record of past events so that we can use data analysis to find recurring patterns. From those patterns, you build predictive models using machine learning algorithms that look for trends and predict future changes (Data exploration).

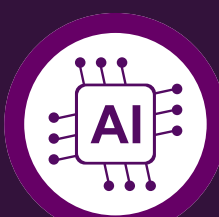


Modeling

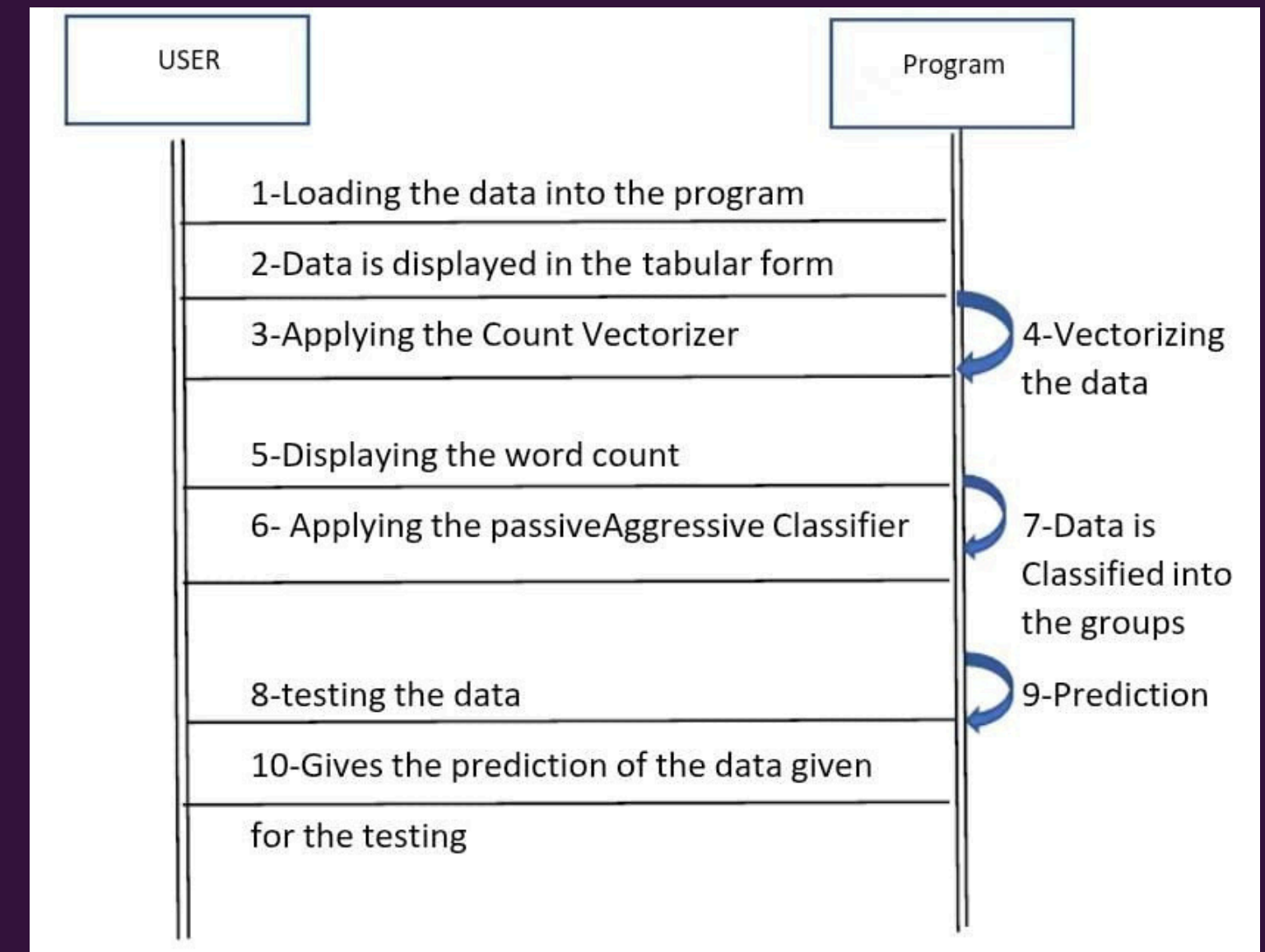
Problem is incomplete without a solution

 Requirements for detecting fake news

 Loading the data sets

 Applying the classifiers

 Checking results



"Do not blame the world,
Find a solution"

Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
MAX_SEQUENCE_LENGTH = 5000
MAX_NUM_WORDS = 25000
EMBEDDING_DIM = 300
TEST_SPLIT = 0.2
```

```
TEXT_DATA = 'https://raw.githubusercontent.com/lutzhamel/
fake-news/master/data/fake_or_real_news.csv'
```

```
from sklearn.metrics import accuracy_score
```

```
def evaluate_model(predict_fun, X_train, y_train, X_test, y_test):
    y_predict_train = predict_fun(X_train)
    train_acc = accuracy_score(y_train, y_predict_train)
```

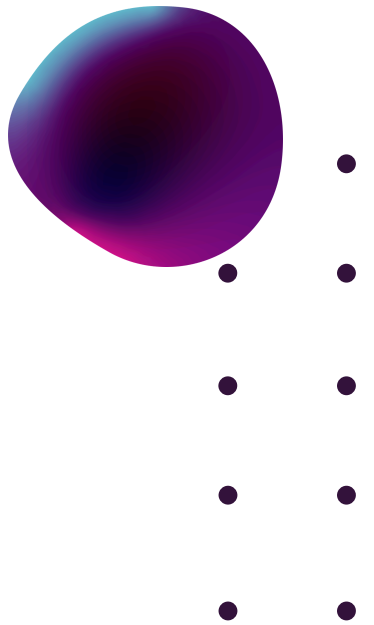
```
y_predict_test = predict_fun(X_test)
test_acc = accuracy_score(y_test, y_predict_test)
```

```
    return train_acc, test_acc
from math import sqrt
def error_conf(error, n):
    term = 1.96*sqrt((error*(1-error))/n)
    lb = error - term
    ub = error + term

    return lb, ub
df = pd.read_csv(TEXT_DATA)
print(df.shape)
print(df.head())
df.drop(labels=['id', 'title'], axis='columns', inplace=True)
# only select stories with lengths gt 0 -- there are some texts with len = 0
mask = list(df['text'].apply(lambda x: len(x) > 0))
df = df[mask]
texts = df['text']
labels = df['label']

print('Found %s texts.' %texts.shape[0])
text_lengths = texts.apply(lambda x: len(x.split(" ")))
plt.hist(text_lengths)
plt.show()

from sklearn.feature_extraction.text import CountVectorizer
vectorizer = CountVectorizer(analyzer = "word",
    binary = True,
    min_df = 2,
    stop_words='english')
docarray = vectorizer.fit_transform(texts).toarray()
docterm = pd.DataFrame(docarray, columns=vectorizer.get_feature_names_out())
```



Evaluation

```
from sklearn.model_selection import train_test_split
docterm_train, docterm_test, y_train, y_test =
train_test_split(docterm, labels, test_size=TEST_SPLIT)
```

```
from sklearn.naive_bayes import MultinomialNB
model = MultinomialNB()
model.fit(docterm_train, y_train)
```

```
train_acc, test_acc = evaluate_model(model.predict,
docterm_train, y_train, docterm_test, y_test)
print("Training Accuracy: {:.2f}%".format(train_acc*100))
print("Testing Accuracy: {:.2f}%".format(test_acc*100))
```

```
n = docterm_test.shape[0]
lb, ub = error_conf(1-test_acc, n)
```

```
print("95% confidence interval: {:.2f}%-{:2f}%".format((1-ub)*100,
(1-lb)*100))
```

accuracy: 0.937.

Confusion matrix, without
normalization

Confusion matrix

Real	955	55
Fake	78	1005
	Real	Fake

Thank you

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**Share your
feedback...!!**

**Guided by :
G Santosh Kumar**

